

SEEHANAH TANG

seehanah@mit.edu ◊ <https://seehanah-tang.github.io/>

EDUCATION

Massachusetts Institute of Technology

Graduating in May 2030

Ph.D in Operations Research

GPA: 5.00/5.00

Brown University

May 2025

Sc.B. in Applied Math-Computer Science

GPA: 4.00/4.00 (Magna Cum Laude), *Honors*

RELEVANT COURSES

Mathematical Programming (15.081)

Robust Optimization (15.094)

Multimodal Machine Learning (6.S985)

Machine Learning under an Optimization Lens (15.095)

Deep Learning and Generative AI (15.099)

Machine Learning (CSCI 1420)

Theory of Computation (BMEVISZA 081)

Graph Theory (BMEVISZA 086)

Real Analysis (MATH 1630/40)

Design and Analysis of Algorithms (CSCI 1570)

Fundamentals of Probability (15.085)

RESEARCH AND INDUSTRY EXPERIENCE

MIT Operations Research Center

Sept 2025 - Present

Graduate Research Assistant for Prof. Dimitris Bertsimas

Cambridge, MA

- Led team to predict positivity of blood culture using multimodal artificial intelligence in collaboration with Hartford Healthcare, integrating clinical and laboratory data and doctor notes to reduce number of blood cultures ordered unnecessarily
- Developing computational methods to detect epistatic genetic interactions (pairwise and higher-order SNP combinations) that provide predictive power beyond additive polygenic risk scores, combining statistical genetics, feature selection, and combinatorial optimization to build a scalable disease-risk prediction pipeline from genome-scale variant data

Brown University School of Public Health

Sept 2024 - May 2025

Honors Thesis Candidate with Prof. Alice Paul

Providence, RI

- Conducted a comparative study on risk score models, including FasterRisk and RiskCD, focusing on optimizing patient outcome predictions, and utilized machine learning techniques to categorize continuous variables for improved model performance
- Designed and executed a simulation study to evaluate computational efficiency and accuracy between the models, validated models with real-world data on tuberculosis diagnosis and treatment adherence to assess clinical applicability and model reliability

The Cigna Group

May 2024 - May 2025

Technology Development Intern

Philadelphia, PA

- Engineered a serverless customer service chatbot and forums app in AWS, utilizing GPT-4 and Retrieval-Augmented Generation (RAG) techniques to enable accurate responses to domain-specific queries

- Incorporated automated testing into pipeline of requester dashboard for Cigna Primary Care Providers (PCP) using Cypress Studio

MIT Sloan School of Management

Jul 2020 - Aug 2024

Research Assistant for Prof. Dimitris Bertsimas

Cambridge, MA

- Led the development of machine learning models using Julia and Python to predict tumor recurrence and optimize treatment strategies for cancers such as IPMN, GIST, CRLM, and soft tissue sarcoma
- Focused on creating interpretable AI models to assist in clinical decision-making, ensuring that the models provide actionable insights for personalized cancer treatments
- Co-author in three peer-reviewed publications in top-tier journals, JAMA Surgery and Lancet Oncology, advancing the application of AI in precision oncology

Benefits Science, MultiPlan

Jun - Aug 2023

Data Science Intern

Boston, MA

- Utilized Databricks and Spark to efficiently process and analyze medical claims data related to hypertension and chronic kidney disease (CKD), implementing data queuing and filtering techniques
- Developed classification models to assess cost-related risk scores for individuals with hypertension and CKD
- Crafted an accessible and user-friendly interface for predictive models on the company's dashboard using React framework Next.js

PUBLICATIONS

"Plasma Proteomics for Unified Multi-Disease Diagnostics and Multi-Horizon Prognostics"

Seehanah Tang, Cindy Beini Wang, Carol Gao, Dewang Agarwal, Joanna G. Kondylis, Dimitris Bertsimas, in preparation, 2026

"Towards Better Clinical Risk Scores with AnnealScore: A Residual-Guided Simulated Annealing Algorithm"

Hannah Eglinton, **Seehanah Tang**, Tongtong Zhao, Yu Yan, Alice Paul, International Journal of Biostatistics, revise and resubmit (submitted 2025, decision received March 2026)

"Early Detection of Blood-Stream Infections via Explainable AI"

Seehanah Tang, Carol Gao, et al., Dimitris Bertsimas, working paper, 2026

"Interpretable artificial intelligence to optimise use of imatinib after resection in patients with localised gastrointestinal stromal tumours: an observational cohort study"

Dimitris Bertsimas, Georgios Antonios Margonis, Suleeporn Sujichantararat, ..., **Seehanah Tang**, et al.

Lancet Oncology, 2024.

"An interpretable AI model for recurrence prediction after surgery in gastrointestinal stromal tumour: an observational cohort study"

Dimitris Bertsimas, Georgios Antonios Margonis, **Seehanah Tang**, et al.

Lancet eClinicalMedicine, 2023.

"Using Artificial Intelligence to Find the Optimal Margin Width in Hepatectomy for Colorectal Cancer Liver Metastases"

Dimitris Bertsimas, Georgios Antonios Margonis, Suleeporn Sujichantararat, ..., **Seehanah Tang**, et al.

JAMA Surgery, 2022.

LEADERSHIP AND COMMUNITY ENGAGEMENT

APMA Connect Program

Oct 2024 - May 2025

Program Lead

Providence, RI

- Interviewed third-year Applied Math concentrators on their experience with the Applied Math department, gathering qualitative data regarding resources, faculty, courses, and classroom experience
- Synthesized feedback and presented to departmental leadership, highlighting opportunities for improved academic support and student engagement

Brown University Orchestra

Sept 2021 - May 2025

First Flutist

Providence, RI

- Performed twice every semester for the Brown University and greater Providence communities, held special programs addressing current events, and participated in composer and artist workshops
- Accompanied world-renowned violinist Itzhak Perlman and acclaimed cellist Zlatomir Fung

The Association of Women in Mathematics (AWM)

Sept 2021 - May 2023

Secretary

Providence, RI

- Led successful initiatives, organizing engaging panels, community events, and mentorship programs, attracting up to 100 participants and fostered the community of women in mathematics at Brown University

TEACHING EXPERIENCE

MIT Sloan School of Management

Sept 2026 - Present

Teaching Assistant for Data, Models, and Decisions

- Held weekly recitations and office hours for a core MBA course of 70+ students, reinforcing lecture material through applied problem-solving
- Authored and graded homework assignments and exams, providing feedback aligned with course learning objectives
- Collaborated with faculty to design assessment materials and calibrate grading standards across the teaching team

Brown University

Jan 2022 - May 2025

Teaching Assistant for Partial Differential Equations, Statistical Inference, Object-Oriented Programming, Probabilistic Models, and Machine Learning

- Conducted informative conceptual and debugging office hours and led weekly lab sections, providing academic support to students
- Evaluated student performance, graded homework and exams, and updated course website material

SKILLS AND LANGUAGES

Technologies: Proficient in Python, Java, R, JavaScript, Julia, Git; Experience with C, MatLab, SQL, Spark

Languages: Native proficiency in English, Chinese; Elementary proficiency in Spanish, Hungarian, ASL